

Traffic Intelligence & Model Evaluation Report

Generated on 2026-09-04 16:16:00 | Platform Version 1.0

1. Executive Summary

This report presents the analysis of **48,187** traffic volume records along the I-94 Metro Interstate corridor. The primary target variable is hourly traffic volume, with an observed average of **3259.62 vehicles/hour** and peak volume reaching **7280 vehicles/hour**. Machine learning models were evaluated on chronological train/val/test splits to ensure zero temporal data leakage.

2. Model Evaluation & Metrics

Model Name	MAE	RMSE	MAPE (%)	R ² Score	Status
Linear Regression	1356.31	1566.99	106.0%	0.3762	Evaluated
Decision Tree	296.46	537.17	14.17%	0.9267	Evaluated
Random Forest	284.67	489.68	13.81%	0.9391	Evaluated
XGBoost	280.12	455.84	14.27%	0.9472	Active Best

3. Selected Production Model

Best Performing Architecture: XGBoost

R² Score: 0.9472 | **MAE:** 280.12 | **RMSE:** 455.84

Latency: 0.0026 ms per prediction.

Deployment Status: Serialized to disk and loaded into active inference pipeline.